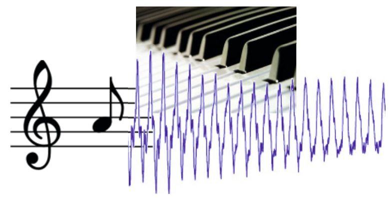
Chapter 1
Music Representations



Music can be represented in many different ways and formats. For example, a composer may write down a composition in the form of a musical score. In a score, musical symbols are used to visually encode notes and how these notes are to be played by a musician. The printed form of a musical score is also referred to as **sheet music**. The original medium of this representation is paper, although it is now also accessible on computer screens through digital images. For electronic instruments and computers, music may be communicated by means of standard protocols such as the widely used Musical Instrument Digital Interface (MIDI) protocol, where event messages specify pitches, velocities, and other parameters to generate the intended sounds. In this book, we use the term **symbolic** to refer to any machine-readable data format that explicitly represents musical entities. These musical entities may range from timed note events, as is the case of MIDI files, to graphical shapes with attached musical meaning, as is the case of music engraving systems. Unlike symbolic representations, audio representations such as WAV or MP3 files do not explicitly specify musical events. These files encode acoustic waves, which are generated when a source (e.g., an instrument) creates a sound that travels to the human ear as air pressure oscillations.

In this book, we distinguish between three main classes of music representations: sheet music, symbolic, and audio. To put it simply, the term **sheet music** stands for visual representations of a score given in printed form or in the form of digitized images. The term **symbolic** comprises any kind of score representation with an explicit encoding of notes or other musical events. Finally, the term **audio** refers to representations of acoustic sound waves. Each of these representations reflects certain aspects of a musical object, but no single representation encompasses all its properties. In this sense, each representation can be considered a projection or a realization of what we generally refer to as a piece of music. In this introductory chapter, we discuss some basic properties of music by means of these different music representations. We start by describing basic elements of Western music notation as used in

M. Müller, *Fundamentals of Music Processing*,
DOI 10.1007/978-3-319-21945-5_1

Fig. 1.1 Sheet music representation of the first five measures of Symphony No. 5 by Ludwig van Beethoven in a piano reduced version.

sheet music representations (Section 1.1). Even though the exact specifications of music notation are not essential in this book, we require basic notions of the pitch, duration, and onset time of musical notes. Then, we summarize basic properties of symbolic representations with a specific focus on MIDI, which is the prevailing standard for controlling music synthesizers (Section 1.2). Finally, we discuss audio representations, which are at the heart of this book. In particular, we deal with aspects concerning the properties of sound waves including frequency, dynamics, and timbre (Section 1.3).

1.1 Sheet Music Representations

Sheet music, also referred to as **musical score**, provides a visual representation of what we commonly refer to—in particular for Western classical music—as the "piece of music." Sheet music describes a musical work using a formal language based on musical symbols and letters, which are depicted in a graphical–textual form. Reading sheet music, a musician can create a performance by following the given instructions. Performing a piece from sheet music, however, not only requires a special form of literacy, i.e., the ability to understand the music notation, but also involves a creative process. A musical score is rarely played mechanically. Musicians may shape the flow of the music by varying the tempo, dynamics, and articulation, thus resulting in a personal interpretation of the given musical score. In this sense, rather than giving rigid specifications, sheet music can be considered as a guide for performing a piece of music leaving room for different interpretations.

As a first example, let us consider Symphony No. 5 in C minor by Ludwig van Beethoven, which is one of the most popular and best-known compositions in classical music. It begins with a short musical idea, the famous "short-short-short-long" **motif**, which is commonly referred to as the "fate motif" of Beethoven's Fifth. Figure 1.1 shows a sheet music representation of the first five measures in a piano reduced version. In the following sections, we explain the meaning of the musical symbols in more detail while introducing some music notations used throughout this book. The Beethoven piece will serve as our running example.

1.1.1 Musical Notes and Pitches

In music, the term **note** is often used in a rather loose way and may refer to both a musical symbol (when talking about score representations) as well as a pitched sound (when talking about audio representations). In this section, we employ the term to refer to musical symbols used in Western music notation. Each note has several attributes that determine the relative duration and the pitch of a sound to be performed by a musician. For example, in the case of a piano, the pitch of a note tells a musician which key is to be pressed on the keyboard, and the duration of the note determines how long this key is to be held. The notion of **pitch** is not strict and refers to a perceptual property that allows a listener to order a sound on a frequency-related scale. As we will discuss in Section 1.3 in more detail, playing a note on an instrument results in a (more or less) periodic sound of a certain fundamental frequency. This fundamental frequency is closely related to what is meant by the pitch of a note. In the following discussion, we use the term “pitch” in an intuitive way. It allows us to order pitched sounds from “lower” to “higher”—similarly to the keys of a piano keyboard ordered from left to right.

Two notes with fundamental frequencies in a ratio equal to any power of two (e.g., half, twice, or four times) are perceived as very similar. Because of that, all notes with this kind of relation can be grouped under the same **pitch class**. This observation also leads to the fundamental notion of an **octave**, which is defined to be the interval between one musical note and another with half or double its fundamental frequency. Using this definition, a pitch class is a set of all pitches or notes that are an integer number of octaves apart.

In order to describe music using a finite number of symbols, one needs to discretize the space of all possible pitches. This leads to the notion of a **musical scale**, which can be thought of as a finite set of representative pitches. Because of the close octave relationship of pitches, scales are generally considered to span a single octave, with higher or lower octaves simply repeating the pattern. A musical scale can then be specified by a division of the octave space into a certain number of scale steps. The elements of a scale are often simply referred to as the **notes** of the scale and are ordered according to their respective pitches.

In music history, many different scales have been suggested and used, and there have been fierce discussions about the suitability of specific scales. The appropriateness of a scale very much depends on the kind of music to be described, the instruments used, the musical genre, or the cultural background. A scale that is suited for representing Western piano music may not be suited for representing Indian sitar music. A scale used for Gregorian chant of the 10th century may not be a good choice for describing experimental music of the 20th century. There is no universally valid musical scale, and the choice of a musical scale necessarily goes along with simplifications typically imposed by practical considerations.

In this book, we only consider the case of the **twelve-tone equal-tempered scale**, where an octave is subdivided into twelve scale steps. The fundamental frequencies of these steps are equally spaced on a logarithmic frequency axis (see Section 1.3.2).

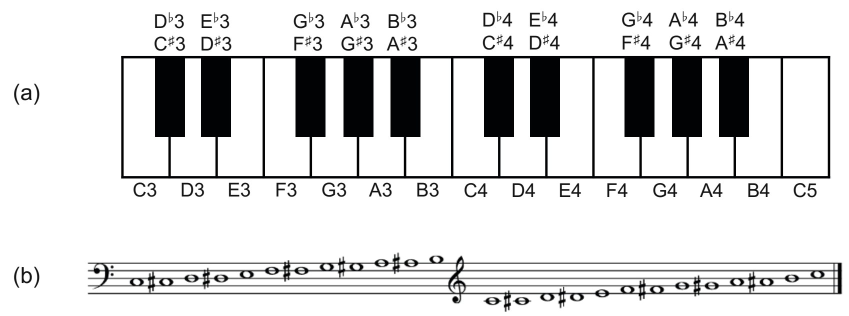


Fig. 1.2 **(a)** Section of piano keyboard with keys ranging from C3 to C5. **(b)** Corresponding notes using Western music notation.

The difference between the fundamental frequencies of two subsequent scale steps is also called a **semitone**, which is the smallest possible interval in this scale.

In the twelve-tone equal-tempered scale, there are twelve pitch classes. In Western music notation, these pitch classes are denoted by combining a letter-name and accidentals. Seven of the pitch classes (corresponding to C major) are denoted by the letters C, D, E, F, G, A, B. These pitch classes correspond to the white keys of a piano keyboard (see Figure 1.2). The remaining five pitch classes correspond to the black keys of a piano keyboard and are denoted by a combination of a letter and an **accidental** ($\sharp$, $\flat$). A sharp ($\sharp$) raises a note by a semitone, and a flat ($\flat$) lowers it by a semitone. The accidentals are written after the note name. For example, $D^\sharp$ represents D-sharp and $D^\flat$ represents D-flat. In the equal-tempered scale, the remaining five pitches can be either denoted by $C^\sharp$, $D^\sharp$, $F^\sharp$, $G^\sharp$, $A^\sharp$ or by $D^\flat$, $E^\flat$, $G^\flat$, $A^\flat$, $B^\flat$. For example, $C^\sharp$ and $D^\flat$ represent the same pitch class,[1] even though from a musical point of view one distinguishes between these two concepts.

To name the notes of the twelve-tone equal-tempered scale, in addition to indicating the pitch class, one needs to provide an identifier for the octave. Following the **Scientific Pitch Notation**, each note is specified by the pitch class name, followed by a number that indicates the octave. The note A4 is determined to have a fundamental frequency of 440 Hz and serves as a reference. The **octave number** increases by one upon an ascension from a note with pitch class B to one with pitch class C. For example, the note B4 is followed by the note C5. Similarly, the octave number decreases by 1 upon a descent from a C to a B. The lowest note C0 in this notation has a fundamental frequency in the region of 16 Hz, which is already below what a human can acoustically perceive. Figure 1.2 shows the notes from C3 to C5 along with the corresponding keys of a piano keyboard.

Ordering all notes of the equal-tempered scale according to their pitches, one obtains an equal-tempered **chromatic scale**, where all notes of the scale are equally spaced. The term **chromatic** is derived from the Greek word **chroma**, meaning color. In the music context, the term "chroma" closely relates to the twelve different

[1] This phenomenon is also known as **enharmonic equivalence**.

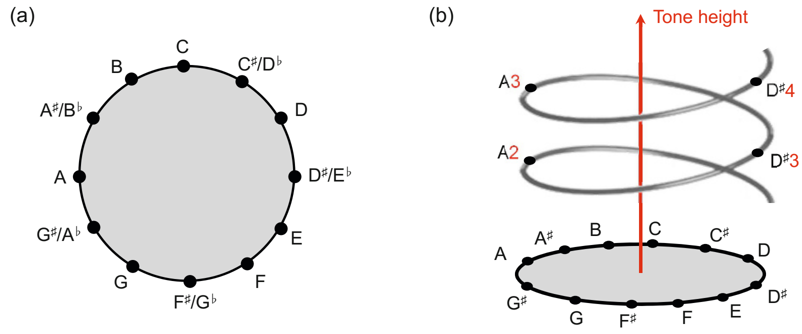


Fig. 1.3 **(a)** Chromatic circle. **(b)** Shepard's helix of pitch [29].

Fig. 1.4 **(a)** Staff. **(b)** Staff with G-clef. **(c)** Staff with F-clef.

pitch classes. For example, the notes C2 and C5 both have the same chroma value C. In other words, all notes that have the same chroma value belong to the same pitch class. Recall that notes that belong to the same pitch class (or have the same chroma value) are perceived as similar in a certain way. In contrast, notes that belong to different pitch classes (or have different chroma values) are perceived as dissimilar. This justifies the usage of the term "chroma" in the sense that notes with different chroma values have a different "sound color." The cyclic nature of chroma values is illustrated by the **chromatic circle** as shown in Figure 1.3a. Extending this notion, **Shepard's helix of pitch** represents the linear pitch space as a helix wrapped around a cylinder so that octave-related pitches lie along a single vertical line [29]. The projection of the cylinder onto the horizontal plane yields the chromatic circle. The factorization of a pitch into a chroma value and an octave number will play an important role in this book. The chroma components of pitches can be used to yield mid-level representations, which turn out to be a powerful tool for various music analysis and retrieval applications.

1.1.2 Western Music Notation

Generally speaking, **music notation** refers to a system for graphically representing music through symbols. The standard Western music notation is based on a **staff**, which is a set of five horizontal lines and four spaces each representing a different

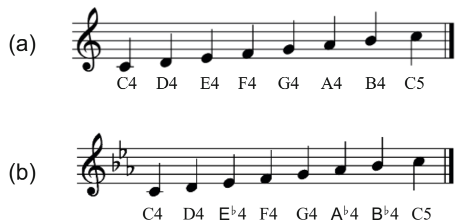


Fig. 1.5 **(a)** Musical score of a C-major scale starting with C4 and ending with C5. **(b)** Key signature consisting of three flats converting the notes into a C-minor scale.

musical pitch (see Figure 1.4a). Appropriate music symbols, depending upon the intended effect, are placed on the staff according to their corresponding pitch or function. Pitch is shown by placement of note symbols on the staff—sometimes modified by accidentals. The higher the placement within a given staff, the higher the pitch of the corresponding note. Furthermore, the duration is indicated by the shapes of the note symbols as well as additional symbols such as dots and ties.

Notation is read from left to right. A staff generally begins with a **clef** symbol, which indicates the position of one particular note on the staff. For example, by convention, the **treble clef** (𝄞), also known as the G-clef, indicates that the second line is the pitch G4 (see Figure 1.4b). Similarly, the **bass clef** (𝄢), also known as the F-clef, indicates that the fourth line is the pitch F3 (see Figure 1.4c). There are also further clef symbols and clef positions. The details are not important in this book. However, one should keep in mind that the clef symbol, along with its position, serves as a reference in relation to which the meaning of the notes positioned on any line or space of the staff can be determined. Notes representing a pitch outside the scope of the five-line staff can be described using **ledger lines**, which provide a single note with additional lines and spaces (see, e.g., the C4 in Figure 1.5).

Following the clef, the **key signature** on a staff indicates the key of the piece by specifying that certain notes are flat or sharp throughout the piece, unless otherwise specified. For example, the notes shown in Figure 1.5a are C4, D4, E4, F4, G4, A4, B4, C5 thus forming a C-major scale. Using the key signature consisting of three flats as shown in Figure 1.5b, the notes become C4, D4, E$^\flat$4, F4, G4, A$^\flat$4, B$^\flat$4, C5 thus forming a (natural) C-minor scale.

Music is typically organized into temporal units, referred to as **beats**. Repeating sequences of stressed and unstressed beats, in turn, form higher temporal patterns, which are related to what is called the **rhythm** of music and is expressed in terms of the musical **meter**. A **measure** (or **bar**) is a segment of time defined by a given number of beats. Dividing music into measures not only reflects it rhythmic nature, but also provides regular reference points within it. In music notation, the temporal structure of a piece is indicated by the **time signature**, which appears in a staff after the key signature. Typically, a time signature consists of two numerals, one stacked above the other. The lower numeral indicates the note duration that represents one beat (given as a fraction with regard to a whole note), while the upper numeral indicates how many such beats are in a measure. For example, the time signature $\substack{6\\8}$ shown in Figure 1.6b indicates that a measure consists of six beats, where a beat has the duration of an eighth note. In sheet music, two subsequent measures are

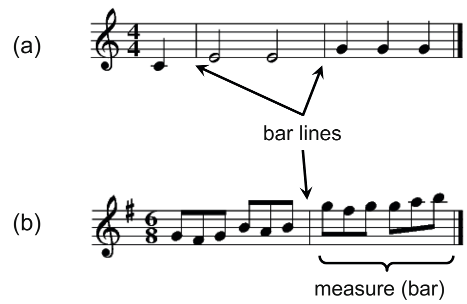


Fig. 1.6 Notation of time signature. **(a)** Four quarter notes per measure with upbeat. **(b)** Six eighth notes per measure.

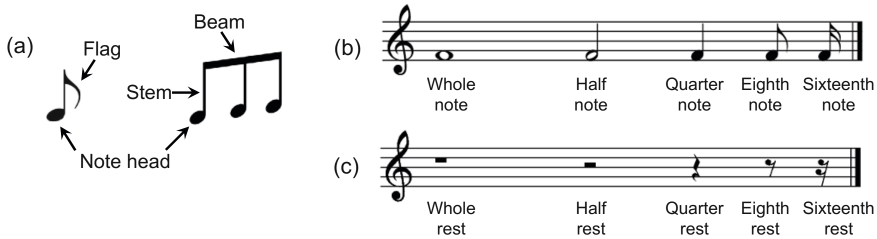


Fig. 1.7 **(a)** Parts of a note. **(b)** Notation for different durations of notes. **(c)** Notation for different durations of rests.

separated by a vertical line drawn through the staff, which are referred to as **bar lines**.

After specifying the clef, the key signature, and the time signature, which all reflect global characteristics of the piece and hold for the entire staff (if not redefined explicitly), the actual notes are specified. As illustrated by Figure 1.7a, each note is represented by a symbol that consists of a **note head** and possibly a **stem** and a **flag**. Sometimes several notes are combined by a **beam**. As discussed above, a note's pitch is indicated by its placement on the staff and possibly by an accidental, where the clef symbol serves as a reference pitch. The duration of a note is defined in a relative fashion by means of its **note value**, which is indicated by the color or shape of the note head, the presence or absence of a stem, and the presence or absence of flags (see Figure 1.7b). The whole note is the reference value, and the other notes are named in accordance. For example, a **half note** has half the length of a whole note, a **quarter note** has a quarter the length of a whole note, and so on. For each note value, there also exists a **rest symbol** of equivalent duration, which expresses an interval of silence in a piece of music (see Figure 1.7c).

The musical onset times of the notes are specified in a relative fashion and follow from the horizontal formation of the note symbols. Notes that are to be played at the same time are given by vertically aligned musical symbols. In this case, different notes may share the same stem and flag as illustrated by Figure 1.1. Once the physical duration of a beat is known, the physical onset times of the notes can be derived from the relative timing. The duration of a beat is given by the **tempo** indication specified in beats per minute (BPM). For example, a specification of 120 BPM

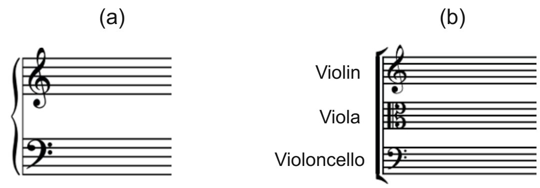


Fig. 1.8 **(a)** Staff system (grand staff) as used for piano. **(b)** Staff system as used for strings.

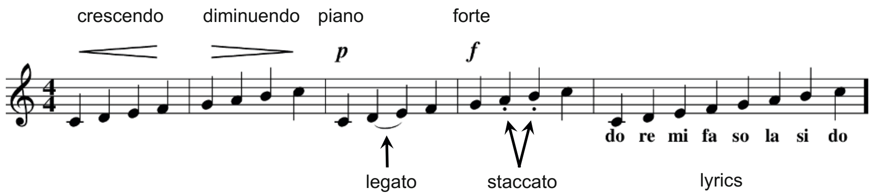


Fig. 1.9 Musical score with various symbols used for indicating dynamics and articulation.

means that 120 beats are to be played within one minute. In the case that a beat corresponds to a quarter note, 120 BPM implies that the duration of a quarter note is half a second. Composers often suggest a tempo notated above the first staff line of the piece. For example, in Figure 1.1, the suggested tempo is 108 BPM with a beat being a half note (𝅗𝅥 = 108). However, when performing a piece, musicians often significantly deviate from the suggested tempo.

To notate music that is played on a piano or is played by different musicians on various instruments, one often uses several staves to notate the various musical voices. A single vertical line drawn to the left of multiple staves creates a **staff system**, which indicates that the music on all the staves is to be played simultaneously. A bracket is an additional vertically aligned symbol joining staves. This symbol shows groupings of instruments that function as a unit, such as the string section of an orchestra (see Figure 1.8b). When music notated across different staves is intended to be played at once by a single performer (usually a keyboard instrument or the harp), a **grand staff** is created by joining the two staves by a brace. For example, in the case of piano music, one has two staves, where the upper staff uses a treble clef and the lower staff uses a bass clef (see Figure 1.8a). When playing the piano, the upper staff is normally played with the right hand and the lower staff with the left hand. This is the case with our Beethoven example shown in Figure 1.1.

Besides the aforementioned attributes, music notation may contain many more instructions to the musician regarding matters such as tempo, dynamics, and expression. For example, the overall **tempo** and **style** of the piece may be specified by textual notations such as *Allegro con brio* (fast with vigor and spirit) or *Andante con moto* (moderate tempo with motion). Other directions such as **accelerando** (gradually becoming faster) or **ritardando** (gradually becoming slower) refer to local tempo deviations. Similarly, **dynamics**, which refers to the volume of a sound or note, may be described by terms such as **forte** (loud), **piano** (soft), **crescendo** (gradually becoming louder), or **diminuendo** (gradually becoming softer). For vo-

Fig. 1.10 Sheet music representation of the full orchestral score of the beginning of Beethoven's Fifth Symphony (from Breitkopf & Härtel, Leipzig, 1862).

cal music, **lyrics** may be written above or below staff lines. Other symbols such as **articulation** marks are used to indicate how certain notes are to be played. For example, a **staccato** mark (a dot placed above or below a note) signifies that a note is to be played with shortened duration detached from the subsequent note, whereas a **legato** mark (a curved line placed above or below a group of notes) indicates that musical notes are played smoothly and connected (see Figure 1.9).

We close this section by coming back to our Beethoven example. In the piano reduced version shown in Figure 1.1, the score shows a system with two staves, where the upper staff for the right hand starts with a G-clef 𝄞 and the lower staff for the left hand with an F-clef 𝄢. Both staves are equipped with a key signature (three flats ♭) and a time signature (two quarter-note beats per measure $\frac{2}{4}$). The score reveals that the first five measures consist of two "short-short-short-long" patterns, where the second fate motif is played lower than the first fate motif. Further instructions are given in the form of additional symbols and textual notations. For example, a fermata sign (𝄐) indicates that the respective note duration should be prolonged. The pedaling information (𝆮 𝆯) tells the musician to hold the sustain pedal, which can have a significant impact on the sound. The overall tempo is indicated by “Allegro con brio” and a metronome specification (108 half notes per minute) (see Exercise 1.1). Finally, the symbol ***ff*** stands for “fortissimo” or “very loud” and indicates the dynamics.

Let us now have a look at Figure 1.10, which shows the full orchestral score of the beginning of Beethoven’s Fifth as used by conductors to direct rehearsals and performances. The shown excerpt is the scanned version of an edition published by Breitkop & Härtel in the year 1862. The various staves of the system specify the music according to the different instruments lined up in a fixed order. From top to bottom, the voices for the woodwinds (flute, oboe, clarinet, bassoon), the brass (French horn, trumpet) and percussion (timpani), and the strings (violin, viola, cello, double bass) are listed. For certain instruments such as the violin, there may be more than one musical voice to be played, each specified by a separate staff (e.g., violin I and violin II).

In this section, we have only scratched the surface of Western music notation. Rather than giving a comprehensive overview, our goal was to build up some intuition while introducing some basic terminology. Furthermore, we wanted to indicate that music notation is far from being comprehensive. Many of the symbols only give a vague description of how the notes should be played leaving room for artistic freedom and creativity. Furthermore, as indicated by the full score and the piano transcript of Beethoven’s Fifth, there may exist different score versions of the same piece of music. For most parts of this book, it suffices to have a rough understanding of musical concepts examined in this chapter. The aspects of pitch and timing will be picked up again when discussing various kinds of derived music representations.

1.2 Symbolic Representations

As discussed at the beginning of this chapter, symbolic representations describe music by means of entities that have an explicit musical meaning and, given in some digital format, can be parsed by a computer. Any kind of digital data format may be regarded as “symbolic” since it is based on a finite alphabet of letters or symbols. For example, the pixels in a digital image file or the samples in a digital audio file may be regarded as symbols or basic entities. However, considering these entities individually, no musical meaning can be inferred. Therefore, neither scanned images nor digitized music recordings are regarded as being symbolic music formats. Similarly, graphical shapes in vector graphics representations are not considered to be musical entities as long as no musically meaningful specification of the shapes is given. Still, there is a wide range of what may be considered as symbolic music. In this section, we discuss some examples including piano-roll representations, MIDI representations, and other symbolic formats that encode sheet music. Furthermore, we touch on optical music recognition (OMR), which is the process of converting digital scans of printed sheet music into symbolic representations.

(a) (b)

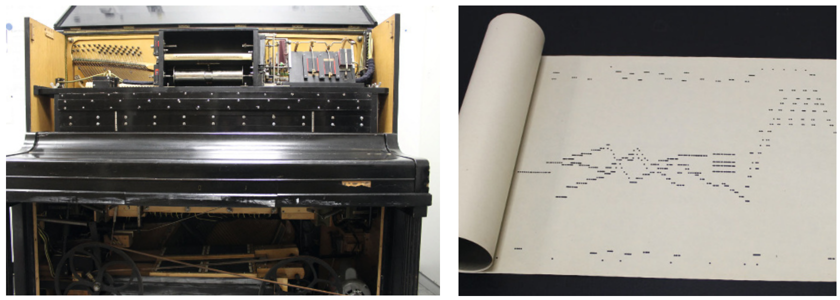


Fig. 1.11 **(a)** Player piano. **(b)** Piano roll. (Reprinted by kind permission of the Institut für Musikwissenschaft der Goethe-Universität Frankfurt)

1.2.1 Piano-Roll Representations

We start with a symbolic representation having a history of more than one hundred years. In the late 19th and early 20th century, self-playing pianos, so-called **player pianos** (see Figure 1.11a), became quite popular with a peak in 1924, before being replaced by phonograph recordings. Player pianos contained pneumatic mechanisms to automatically operate the key and pedal movements according to the instructions specified by a prestored piano-roll medium. A **piano roll** is a continuous roll of paper with perforations (holes) punched into it. The perforations represent note control data (see Figure 1.11b). The roll moves over a reading system known as a tracker bar, and the playing cycle for each musical note is triggered when a perforation crosses the bar and is read. Rolls for player pianos were generally made from recorded performances of musicians. This way, the playing of many famous pianists and composers including Gustav Mahler, Edvard Grieg, Scott Joplin, or George Gershwin is preserved on piano rolls. Typically, a pianist would sit at a specially designed player piano, and the pitch and duration of any notes played would be perforated into a blank roll, together with the duration of the sustain and soft pedal. Player pianos can also recreate the dynamics of a pianist's performance by means of specially encoded control perforations placed towards the edges of a music roll.

In the following, a **piano-roll representation** is understood to be a geometric visualization of the note information as specified by a piano roll. The horizontal axis of this two-dimensional representation encodes time, whereas the vertical axis encodes pitch. Every note is described by an axis-parallel rectangle coding three parameters. The first parameter is the onset time, given by the leftmost horizontal coordinate of the rectangle, and the second is the pitch, given by the lower vertical coordinate of the rectangle. Finally, the third parameter is the duration of the note, encoded by the width of the rectangle.

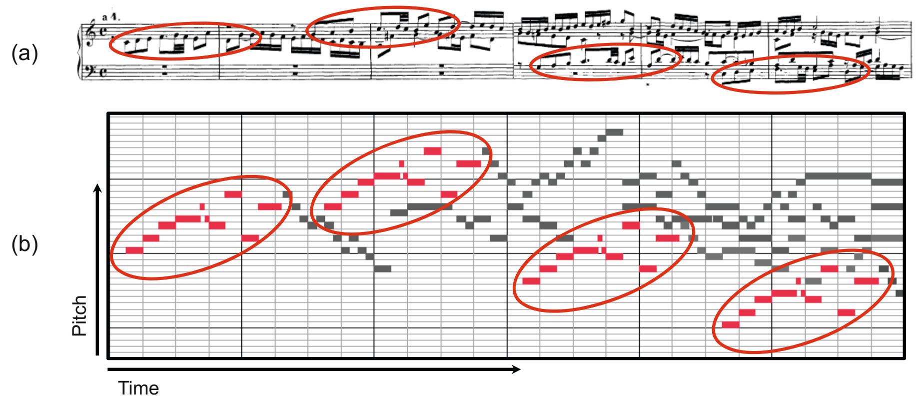


Fig. 1.12 **(a)** Sheet music representation and **(b)** piano-roll representation of the beginning of Fugue BWV 846 in C major by Johann Sebastian Bach. The four occurrences of the theme are highlighted.

Figure 1.12b shows a piano-roll representation of the beginning of Fugue BWV 846 in C major by Johann Sebastian Bach. For comparison, Figure 1.12a shows the corresponding part in a sheet music representation. Generally, a **fugue** is a compositional technique using two or more musical voices, built on a musical theme (or subject) that is introduced at the beginning by one voice and then repeated at different pitches in the other voices. Fugue BWV 846 consists of four voices. Although played on a keyboard instrument, the four voices are referred to as soprano (highest voice), alto (second highest voice), tenor (third highest voice), and bass (lowest voice). The fugue starts with the main theme in the alto, which is then repeated in the soprano, the tenor, and finally in the bass. As shown by Figure 1.12, the four occurrences of the theme are hard to detect in the sheet music representation, but can be easily seen in the piano-roll representation, where each one corresponds to a pattern shifted in the time–pitch plane.

While they are a considerable simplification of what is notated in sheet music, piano-roll representations visually describe the most important attributes of musical notes in an easy-to-understand way. Therefore, we will often use piano-roll representations when describing and talking about symbolic music. Furthermore, as we will see in later chapters, one can also derive similar representations from other music encodings including MIDI and audio. In this sense, piano rolls can be seen as a kind of **mid-level representation** on the basis of which semantic relations can be established across various manifestations of music.

1.2.2 MIDI Representations

The next symbolic representation we want to discuss is based on the **MIDI** standard, which stands for **Musical Instrument Digital Interface**. Although MIDI was not originally developed to be used as a symbolic music format and imposes many limitations on what can actually be represented, the importance of MIDI is due to its widespread usage over the last three decades, and the abundance of MIDI data freely available on the web. From a music encoding point of view, one needs to keep in mind that the quality of available MIDI data is sometimes questionable.

MIDI was originally developed as an industry standard to get digital electronic musical instruments from different manufacturers to work and play together. It was the advent of MIDI in 1981–1983 that caused a rapid growth of the electronic musical instrument market. MIDI allows a musician to remotely and automatically control an electronic instrument or a digital synthesizer in real time. As an example, let us consider a digital piano, where a musician pushes down a key of the piano keyboard to start a sound. The intensity of the sound is controlled by the velocity of the keystroke. Releasing the key stops the sound. Instead of physically pushing and releasing the piano key, the musician may also trigger the instrument to produce the same sound by transmitting suitable MIDI messages, which encode the note-on, the velocity, the note-off, and other information. These MIDI messages may be automatically generated by some other electronic instrument or may be provided by a computer. It is an important fact that MIDI does not represent musical sound directly, but only represents performance information encoding the instructions about how an instrument has been played or how music is to be produced.

The original MIDI standard was later augmented to include the **Standard MIDI File** (SMF) specification, which describes how MIDI data should be stored on a computer. In the following, we denote SMF files simply as **MIDI files** or **MIDI representations**. The SMF file format allows users to exchange MIDI data regardless of the computer operating system and has provided a basis for an efficient internet-wide distribution of music data, including numerous websites devoted to the sale and exchange of music. A MIDI file contains a list of MIDI messages together with timestamps, which are required to determine the timing of the messages. Further information (called meta messages) is relevant to software that processes MIDI files.

For our purposes, the most important MIDI messages are the note-on and the note-off commands, which correspond to the start and the end of a note, respectively. Each note-on and note-off message is, among others, equipped with a MIDI note number, a value for the key velocity, a channel specification, as well as a timestamp. The **MIDI note number** is an integer between 0 and 127 and encodes a note's pitch. Here, MIDI pitches are based on the equal-tempered scale as discussed in Section 1.1.1. Similarly to an acoustic piano, where the 88 keys of the keyboard correspond to the musical pitches A0 to C8, the MIDI note numbers encode, in increasing order, the musical pitches C0 to G$^\sharp$9. For example, note C4 has the MIDI note number 60, whereas the concert pitch A4 has the MIDI note number 69.

The **key velocity** is again an integer between 0 and 127, which controls the intensity of the sound—in the case of a note-on event it determines the volume, whereas

(a)

(b)

Time (Ticks)	Message	Channel	Note Number	Velocity
60	NOTE ON	1	67	100
0	NOTE ON	1	55	100
0	NOTE ON	2	43	100
55	NOTE OFF	1	67	0
0	NOTE OFF	1	55	0
0	NOTE OFF	2	43	0
5	NOTE ON	1	67	100
0	NOTE ON	1	55	100
0	NOTE ON	2	43	100
55	NOTE OFF	1	67	0
0	NOTE OFF	1	55	0
0	NOTE OFF	2	43	0
5	NOTE ON	1	67	100
0	NOTE ON	1	55	100
0	NOTE ON	2	43	100
55	NOTE OFF	1	67	0
0	NOTE OFF	1	55	0
0	NOTE OFF	2	43	0
5	NOTE ON	1	63	100
0	NOTE ON	2	51	100
0	NOTE ON	2	39	100
240	NOTE OFF	1	63	0
0	NOTE OFF	2	51	0
0	NOTE OFF	2	39	0

(c)

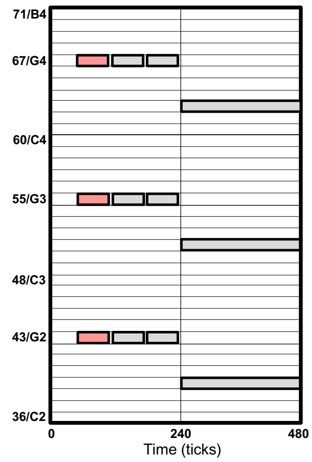


Fig. 1.13 Various symbolic music representations of the first twelve notes of Beethoven's Fifth. **(a)** Sheet music representation. **(b)** MIDI representation (in a simplified, tabular form). **(c)** Piano-roll representation.

in the case of a note-off event it controls the decay during the release phase of the tone. The exact interpretation of the key velocity, however, depends on the respective instrument or synthesizer. The **MIDI channel** is an integer between 0 and 15. Intuitively speaking, this number prompts the synthesizer to use the instrument that has been previously assigned to the respective channel number. Note that each channel, in turn, supports polyphony, i.e., multiple simultaneous notes. Finally, the **time stamp** is an integer value that represents how many clock pulses or **ticks** to wait before the respective note-on or note-off command is executed. Before we comment in more detail on the timing concept employed by MIDI, we illustrate the MIDI representation by means of our Beethoven example. Figure 1.13b shows a (simplified and tabular) MIDI encoding of the first fate motif corresponding to the twelve notes of the score in Figure 1.13a. In this example, the notes of the right hand are assigned to channel 1 and the notes of the left hand to channel 2. The notes specified by corresponding note-on and note-off events in the MIDI file can also be visualized by a piano-roll representation (see Figure 1.13c). In case we are only interested in the note events (and not the channel and velocity information), this is how we represent MIDI information.

An important feature of the MIDI format is that it can handle musical as well as physical onset times and note durations. Similarly to sheet music representations, MIDI can express timing information in terms of musical entities rather than using absolute time units such as microseconds. To this end, MIDI subdivides a quarter note into basic time units referred to as **clock pulses** or **ticks**. The number of pulses per quarter note (PPQN) is to be specified at the beginning, in the so-called **header** of a MIDI file, and refers to all subsequent MIDI messages. A common value is

120 PPQN, which determines the resolution of the time stamps associated to note events. As mentioned above, a time stamp indicates how many ticks to wait before a certain MIDI message is executed, relative to the previous MIDI message. For example, the first note-on message with MIDI note number 67 is executed after 60 ticks, corresponding to the eighth rest at the beginning of Beethoven's Fifth. The second and third note-on messages are executed at the same time as the first one, encoded by the tick value zero. Then, after 55 ticks, MIDI note 67 is switched off by the note-off message and so on.

Like the sheet music representation, MIDI also allows for encoding and storing absolute timing information, however, at a much finer resolution level and in a more flexible way. To this end, one can include additional tempo messages that specify the number of microseconds per quarter note. From the tempo message, one can compute the absolute duration of a tick. For example, having 600000 μs per quarter note and 120 PPQN, each tick corresponds to 5000 μs. Furthermore, one can derive from the tempo message the number of quarter notes played in a minute, which yields the tempo measured in **beats per minute** (BPM). For example, the 600000 μs per quarter note correspond to 100 BPM. While the number of pulses per quarter note is fixed throughout a MIDI file, the absolute tempo information may be changed by inserting a tempo message between any two note-on or other MIDI messages. This makes it possible to account not only for global tempo information but also for local tempo changes such as accelerandi, ritardandi, or fermate.

In this section, we have briefly touched on MIDI and its functionality. As noted above, MIDI was originally designed to solve problems in electronic music performance and is limited in terms of the musical aspects it represents. For example, MIDI is not capable of distinguishing between a D$^\sharp$4 and an E$^\flat$4, both of which have the MIDI note number 63. Also, information on the representation of beams, stem directions, or clefs is not encoded by MIDI. Furthermore, MIDI does not define a note element explicitly; rather, notes are bounded by note-on and note-off events (or note-on events with velocity 0). Rests are not represented at all and must be inferred from the absence of notes.

1.2.3 Score Representations

Within the class of symbolic music representations, we want to distinguish one subclass we refer to as **score representations**. A representation from this subclass is defined to yield explicit information about musical symbols such as the staff system, clefs, time signatures, notes, rests, accidentals, and dynamics. In this sense, score representations are, compared with MIDI representations, much closer to what is actually shown in sheet music. For example, in a score representation, the notes D$^\sharp$4 and E$^\flat$4 would be distinguishable, and the musical onset times are specified. However, a score representation may not contain a description of the final layout and the particular shape of the musical symbols. The process of generating or rendering visually pleasing sheet music representations from score representations is an

Fig. 1.14 Different sheet music representations corresponding to the same score representation of the beginning of Prelude BWV 846 (C major) by Johann Sebastian Bach. From top left to bottom right, a computer-generated, a handwritten, and two traditionally engraved representations are shown.

art in itself. In former days, the art of drawing high-quality music notation for mechanical reproduction was called **music engraving**. Nowadays, computer software or **scorewriters** have been designed for the purpose of writing, editing, and printing music, though only a few produce results comparable to high-quality traditional engraving. Figure 1.14 illustrates this by showing different sheet music representations corresponding to the same score.

In this book, we do not give an overview of existing symbolic score formats. Instead, as an example, we discuss some aspects of **MusicXML**, which has been developed to serve as a universal format for storing music files and sharing them between different music notation applications. Following the general XML (Extensible Markup Language) paradigm, MusicXML is a textual data format that defines a set of rules for encoding documents in a way that is both human and machine readable. For example, Figure 1.15 shows how a note $E^\flat 4$ is encoded. In the MusicXML encoding of the half note $E^\flat 4$, the tags `<note>` and `</note>` mark the beginning and the end of a MusicXML note element. The pitch element, delimited by the tags `<pitch>` and `</pitch>`, consists of a pitch class element E (denoting the letter name of the pitch), the alter element -1 (changing E to E flat), and the octave element 4 (fixing the octave). Thus, the resulting note is an $E^\flat 4$. The element `<duration>2</duration>` encodes the duration of the note measured in quarter notes. Finally, the element `<type>half</type>` tells us how this note is actually depicted in the rendered sheet music.

There are various ways to generate digital score representations. For example, one could manually input the score information in a format such as MusicXML. This, however, is a tedious and error-prone procedure. Music notation software or scorewriters support users in the task of writing and editing digitized sheet music. Such software allows a user to conveniently input and modify note objects by standard computer input devices or electronic keyboards. In the next section, we discuss another way for generating score representations from scanned images of printed sheet music, which is, in a sense, the inverse of a rendering process.

```
<note>
  <pitch>
    <step>E</step>
    <alter>-1</alter>
    <octave>4</octave>
  </pitch>
  <duration>2</duration>
  <type>half</type>
</note>
```

Fig. 1.15 Textual description in the MusicXML format of a half note $E^{\flat}4$. The clef, key signature, and time signature are defined at the beginning of the MusicXML file.

1.2.4 Optical Music Recognition

Sheet music is widely available, and many people are trained to use music notation for studying and playing music. For centuries, music has been documented, transmitted, and distributed in the form of printed sheet music. Music libraries and archives possess huge collections comprising millions of sheet music books, which are now successively being transferred into the digital domain using scanning devices. A digital image resulting from such a scanning process consists of a number of rows and columns of pixels, each pixel encoding the color at a specific point of the scanned page. In other words, a digital image of a sheet music page is by itself a mere accumulation of colored (often black and white) pixels without expressing any deeper musical meaning.

The process of converting digital images of sheet music into symbolic music representations such as MIDI or MusicXML is commonly referred to as **optical music recognition** (OMR).[2] During this process, the image pixels have to be suitably grouped and interpreted in terms of musical symbols. This process is not easy, because of the many ways musical symbols may be engraved into sheet music. As discussed in the last section and illustrated by Figure 1.14, there may be substantial variations in the layout of the symbols and the staff system. Symbols do not always look exactly the same across different editions and may also be degraded in quality by artifacts of the printing or scanning process. Furthermore, musical symbols often intersect with staff lines, and several symbols may be stacked and combined (e.g., several notes sharing the same stem or combined with a beam). As a result, musical scores and the interrelations between musical symbols can become quite complex.

Correctly recognizing and interpreting the meaning of all the musical symbols is easy for a trained human, but hard for a computer. Figure 1.16a shows some examples of typical errors produced by automated OMR procedures. Some of these errors such as missing notes, flags or beams are of local nature, while other errors, such as an incorrectly detected key signature, affect all notes of a staff line. Even worse is the presence of a **transposing instrument**, whose music is notated at a pitch different from the pitch that is actually played (see Figure 1.16c). For example, a clarinet in $B^{\flat}$ is a transposed instrument, where a C in a score sounds like a $B^{\flat}$. Missing this information, which is encoded in textual form in front of a staff line, leads to a mis-

[2] The equivalent in the text domain is known as **optical character recognition** (OCR) with the goal of converting scanned images of printed text into machine-encoded text.

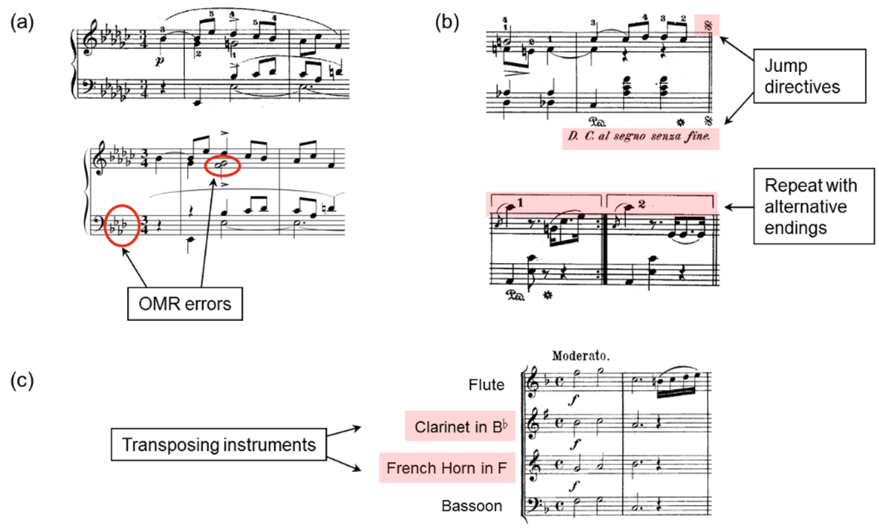


Fig. 1.16 **(a)** Examples of typical OMR errors (top: original score; bottom: OMR result). **(b)** Jump directives and repeats often not detected by OMR. **(c)** Transposed instruments often not interpreted correctly by OMR.

representation of all the notes' pitches. A score may also contain repeat signs with alternative endings or textual jump directives as shown in Figure 1.16b. This information is required to derive the correct sequence of measures to be performed by a musician. Consequently, an error in detecting jump directives may lead to structural misinterpretations of the score. Another problem is that even small artifacts in the scan may lead to confusion with musical symbols, e.g., a small dot being mixed up with a staccato mark. Even though current OMR software is reported to yield highly accurate results, manual postprocessing still seems necessary to obtain high-quality symbolic representation.

1.3 Audio Representation

Music is much more than a symbolic description of the notes to be played. Music is about making, creating, and shaping sounds. When musicians start delving into the music, the playing instructions recede into the background. The musical meter turns into a rhythmic flow, the different note objects melt into harmonic sounds and smooth melody lines, and the instruments communicate with each other. Musicians get emotionally involved with their music and react to it by continuously adapting tempo, dynamics, and articulation. Instead of playing mechanically, they speed up at some points and slow down at others in order to shape a piece of music. Similarly, they continuously change the sound intensity and stress certain notes. All of this results in a unique performance or an interpretation of the piece of music.

From a physical point of view, performing music results in **sounds** or **acoustic waves**, which are transmitted through the air as pressure oscillations. The term **audio** is used to refer to the transmission, reception, or reproduction of sounds that lie within the limits of human hearing. An **audio signal** is a representation of sound. As opposed to sheet music and symbolic representations, an audio representation encodes all information needed to reproduce an acoustic realization of a piece of music. This includes the temporal, dynamic, and tonal microdeviations that make up the specific performance style of a musician. However, in an audio representation, note parameters such as onset times, pitches or note durations are not given explicitly. This makes the analysis and comparison of music signals a difficult task, in particular with regard to polyphonic music, where different instruments and voices are superimposed upon each other. Furthermore, the perception of sounds does not only depend on objective properties of the acoustic wave, but also on subjective criteria as a result of the complex processing a sound undergoes by both the human ear and the brain. The study of subjective human sound perception is called **psychoacoustics**—for further details see [5, 18]. In this section, after having a look at waves and waveforms, we summarize the most important properties of audio representations: frequency and pitch, dynamics, intensity and loudness, as well as timbre.

1.3.1 Waves and Waveforms

A **sound** is generated by a vibrating object such as the vocal cords of a singer, the string and soundboard of a violin, the diaphragm of a kettledrum, or the prongs of a tuning fork. These vibrations cause displacements and oscillations of air molecules, resulting in local regions of compression and rarefaction. The alternating pressure travels through the air as a **wave**, from its source to a listener or a microphone. At its destination, it can then be perceived as sound by the human or converted into an electrical signal by a microphone (see Figure 1.17). In the case of a listener, the outer part of the ear captures the sound wave and passes it to the eardrum, which in turn starts vibrating according to the pressure oscillations. After further processing in the middle and inner ear, the sound wave is transformed into nerve impulses, which are finally sent to and interpreted by the brain. Graphically, the change in air pressure at a certain location can be represented by a **pressure–time plot**, also referred to as the **waveform** of the sound. The waveform shows the deviation of the air pressure from the average air pressure. Figure 1.18 shows a waveform representation of a recording of Beethoven's Fifth Symphony.

In general terms, a (mechanical) **wave** can be described as an oscillation that travels through space, where energy is transferred from one point to another. When a wave travels through some medium, the substance of this medium is temporarily deformed. As described above, sound waves propagate via air molecules colliding with their neighbors. After air molecules collide, they bounce away from each other

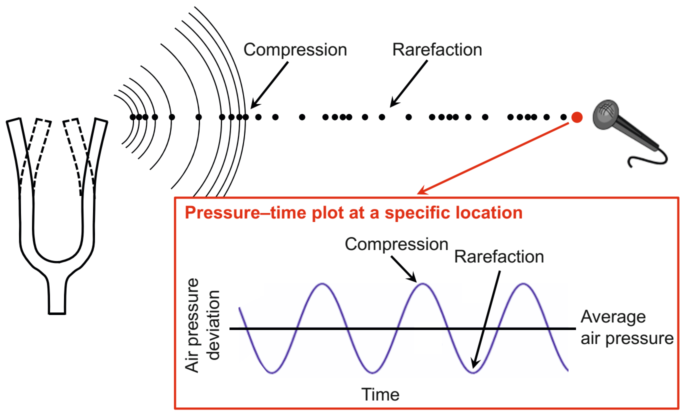


Fig. 1.17 Vibrating tuning fork resulting in a back and forth vibration of the surrounding air particles. The pressure oscillation propagates as a longitudinal wave through the air. The waveform shows the deviation over time of the air pressure from the average air pressure at a specific location (as indicated by the microphone).

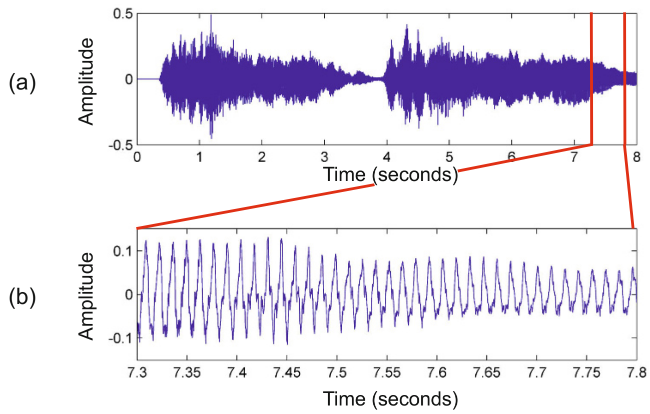


Fig. 1.18 **(a)** Waveform of the first eight seconds of a recording of the first five measures of Beethoven's Fifth as indicated by Figure 1.1. **(b)** Enlargement of the section between 7.3 and 7.8 seconds.

(a restoring force). This keeps the molecules from continuing to travel in the direction of the wave. Instead, they oscillate around almost fixed locations. A general wave can be **transverse** or **longitudinal**, depending on the direction of its oscillation. Transverse waves occur when a disturbance creates oscillations perpendicular (at right angles) to the propagation (the direction of energy transfer). Longitudinal waves occur when the oscillations are parallel to the direction of propagation. According to this definition, a vibration in a string is an example of a transverse wave, whereas a sound wave has the form of a longitudinal wave. A transverse wave can

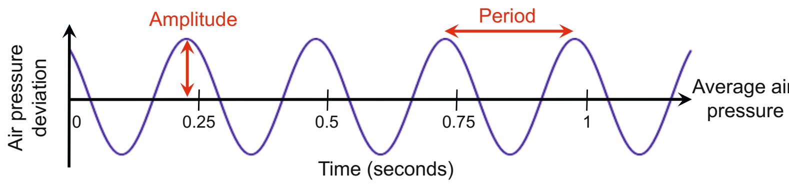


Fig. 1.19 Waveform of a sinusoid with a frequency of 4 Hz.

in fact generate a longitudinal wave and visa versa. An instrument's vibrating string, which oscillates between the two fixed end points, gradually emits its energy to the air, generating a longitudinal sound wave. If this wave, in turn, hits an eardrum, again a transverse wave is generated.

1.3.2 Frequency and Pitch

We have seen that a sound wave can be visually represented by a waveform. If the points of high and low air pressure repeat in an alternating and regular fashion, the resulting waveform is called **periodic**. In this case, the **period** of the wave is defined as the time required to complete a cycle. The **frequency**, measured in **Hertz** (Hz), is the reciprocal of the period. Figure 1.19 shows a **sinusoid**, which is the simplest type of periodic waveform. In this example, the waveform has a period of a quarter second and hence a frequency of 4 Hz. A sinusoid is completely specified by its frequency, its **amplitude** (the peak deviation of the sinusoid from its mean), and its **phase** (determining where in its cycle the sinusoid is at time zero). These three attributes of a sinusoid will become important when analyzing general audio signals (see Section 2.3).

The higher the frequency of a sinusoidal wave, the higher it sounds. The audible frequency range for humans is between about 20 Hz and 20,000 Hz (20 kHz). Other species have different hearing ranges. For example, the top end of a dog's hearing range is about 45 kHz, a cat's is 64 kHz, while bats can even detect frequencies beyond 100 kHz. This is why one can use a dog whistle, which emits **ultrasonic sound** beyond the human hearing capability, to train and to command animals without disturbing nearby people.

The sinusoid can be considered the prototype of an acoustic realization of a musical note. Sometimes the sound resulting from a sinusoid is called a **harmonic sound** or **pure tone**. As indicated in Section 1.1.1, the notion of frequency is closely related to what determines the **pitch** of a sound. In general, pitch is a subjective attribute of sound. In the case of complex sound mixtures its relation to frequency can be especially ambiguous. In the case of pure tones, however, the relation between frequency and pitch is clear. For example, a sinusoid having a frequency of 440 Hz corresponds to the pitch A4. This particular pitch is known as **concert pitch**, and it is used as the reference pitch to which a group of musical instruments are tuned for a performance. Since a slight change in frequency does not necessarily lead to a perceived change, one usually associates an entire range of frequencies with a single pitch.

As mentioned in Section 1.1.1, two frequencies are perceived as similar if they differ by a power of two, which has motivated the notion of an octave. For example, the pitches A3 (220 Hz), A4 (440 Hz), and A5 (880 Hz) sound similar. Furthermore, the perceived distance between the pitches A3 and A4 is the same as the perceived distance between the pitches A4 and A5. In other words, the human perception of pitch is logarithmic in nature. This perceptual property has already been used in Section 1.1.1 when defining the equal-tempered scale that subdivides an octave into twelve semitones based on a logarithmic frequency axis. More formally, using the MIDI note numbers introduced in Section 1.2.2, we can associate to each pitch $p \in [0:127]$ a **center frequency** $F_{\mathrm{pitch}}(p)$ (measured in Hz) by

$$F_{\mathrm{pitch}}(p) = 2^{(p-69)/12} \cdot 440. \tag{1.1}$$

Indeed, this formula yields the frequency $F_{\mathrm{pitch}}(p) = 440$ for the reference pitch $p = 69$ (A4). Increasing the pitch number by 12 (an octave) leads to an increase by a factor of two, i.e., $F_{\mathrm{pitch}}(p+12) = 2 \cdot F_{\mathrm{pitch}}(p)$. Similarly, it is easy to show that the frequency ratio

$$F_{\mathrm{pitch}}(p+1)/F_{\mathrm{pitch}}(p) = 2^{1/12} \approx 1.059463 \tag{1.2}$$

of two subsequent pitches $p+1$ and p is constant (see Exercise 1.6). In other words, multiplying the center frequency of an arbitrary pitch by this constant, the pitch is raised by a semitone. Generalizing the notion of semitones, the **cent** denotes a logarithmic unit of measure used for musical intervals. By definition, an octave is divided into 1200 cents, so that each semitone corresponds to 100 cents. Again the ratio of frequencies one cent apart is constant, yielding the value

$$2^{1/1200} \approx 1.0005777895. \tag{1.3}$$

The difference in cents between two frequencies, say ω_1 and ω_2, is given by

$$\log_2\left(\frac{\omega_1}{\omega_2}\right) \cdot 1200. \tag{1.4}$$

The interval of one cent is much too small to be heard between successive notes. The threshold of what is perceptible, also called the **just noticeable difference**, varies from person to person and depends on other aspects such as the timbre (Section 1.3.4) and the musical context. As a rule of thumb, normal adults are able to recognize pitch differences as small as 25 cents very reliably, with differences of 10 cents being recognizable only by trained listeners.

Real-world sounds are far from being a simple pure tone with a well-defined frequency. Playing a single note on an instrument may result in a complex sound that contains a mixture of different frequencies changing over time. Intuitively, such a **musical tone** can be described as a superposition of pure tones or sinusoids, each with its own frequency of vibration, amplitude, and phase. A **partial** is any of the sinusoids by which a musical tone is described. The frequency of the lowest partial present is called the **fundamental frequency** of the sound. The pitch of a musical tone is usually determined by the fundamental frequency, which is the one created by vibration over the full length of a string or air column of an instrument. A **harmonic** (or a **harmonic partial**) is a partial that is an integer multiple of the fundamental frequency. Partials, as well as harmonics, are counted upwards along the frequency axis. This convention implies that the fundamental frequency is the first partial, as well as the first harmonic of a musical tone. The term **inharmonicity** is used to denote a measure of the deviation of a partial from the closest ideal harmonic, typically measured in cents for each partial. Finally, another term that is often used in music theory is the **overtone**, which is any partial except the lowest. This can lead to numbering confusion when comparing overtones with partials, since the first overtone is the second partial.

Most pitched instruments are designed to have partials that are close to being harmonics, with very low inharmonicity. Thus, for simplicity, one often speaks of the partials in those instruments' sounds as harmonics, even if they have some inharmonicity. Other pitched instruments, especially certain percussion instruments, such as the marimba, vibraphone, bells, and kettledrums (timpani), contain nonharmonic partials, yet give the ear a good sense of pitch. Nonpitched, or indefinite-pitched, instruments, such as cymbals, gongs, or tam-tams, make sounds rich in inharmonic partials. As an example of a harmonic sound, Figure 1.18 shows in its lower part an enlargement of the waveform of the section between 7.3 and 7.8 seconds, which reveals the almost periodic nature of the sound signal. The waveform within these 500 ms corresponds to the sound of a decaying D, which is played by the orchestra in unison in the fourth and fifth measure (see Figure 1.1). Indeed, one counts 37 periods within this section, corresponding to a frequency of 74 Hz—the fundamental frequency of D2.

We close this section on frequency and pitch by looking at harmonics in terms of musical pitches. Let ω denote the center frequency of a musical note, e.g., $\omega = 65.4$ Hz for C2 (having MIDI note number $p = 36$). The harmonic series is an arithmetic series ω, 2ω, 3ω, 4ω, $\ldots$, where the difference between consecutive harmonics is constant and equal to the fundamental. Since our perception of pitch is logarithmic in frequency, we perceive higher harmonics as "closer together" than lower ones. On the other hand, the octave series is a geometric progression ω, 2ω, 4ω, 8ω, $\ldots$, and we hear these distances as "the same" in the sense of musical interval. Consequently, in terms of what we hear, each octave in the harmonic series is divided into increasingly "smaller" and more numerous intervals. In our example, the second harmonic (2ω) sounds like a C3 (one octave higher), the third harmonic (3ω) like a G3 (a so-called **perfect fifth** above C3), and the fourth harmonic (4ω) like a C4 (two octaves higher). Starting with a C2, Figure 1.20 shows for each of the first 16 harmonics the musical note that is closest in terms of the difference between the harmonic's frequency and the center frequency of the note as specified

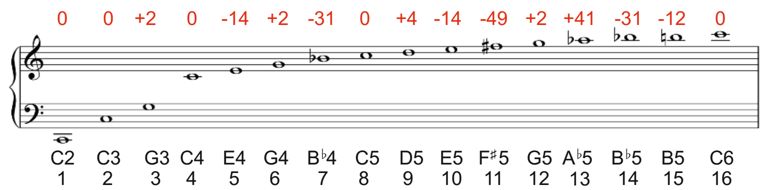


Fig. 1.20 Illustration of the harmonic series in music notation. Starting with the note C2, for each of the first 16 harmonics the closest musical note is shown. On top, the difference (in cents) between a harmonic's frequency and the center frequency of the closest note is shown.

in (1.1) (see also Exercise 1.9). For example, the frequency of the third harmonic is just 2 cents above the center frequency of G3, which is much smaller than the just noticeable difference. In contrast, the frequency of the 11^{th} harmonic is 49 cents below the center frequency of the note F♯5, which is nearly half a semitone and clearly audible. If the harmonics are transposed into the span of one octave (by suitably multiplying or dividing the frequencies by a power of two), they approximate certain notes of the twelve-tone equal-tempered scale. Some of the twelve scale steps are approximated well such as the ones for C (1^{st} harmonic), G (3^{rd} harmonic), or D (9^{th} harmonic), whereas others are problematic such as F♯ (11^{th} harmonic), A♭ (13^{th} harmonic), or B♭ (7^{th} harmonic).

1.3.3 Dynamics, Intensity, and Loudness

As mentioned in Section 1.1.2, a further important property of music concerns the **dynamics**, a general term that is used to refer to the volume of a sound as well as to the musical symbols that indicate the volume. For example, a piano (notated as ***p***) indicates that notes are to be played softly, whereas a forte (notated as ***f***) indicates that notes are to be played loudly. There are many more indicators for describing the dynamics of notes in sheet music. On the audio side, dynamics correlate with a perceptual property called **loudness**, by which sounds can be ordered on a scale extending from quiet to loud. Similarly to the relation between pitch and frequency, loudness is a subjective measure which correlates to objective measures of **sound intensity** and **sound power**. However, loudness also depends on other sound characteristics such as duration or frequency. We will come back to some of these subjective phenomena after having a closer look at the objective measures.

From a physical point of view, it is not easy to strictly define the intensity or power of a sound. In the following, we only give some intuitive explanations. In general, **power** is the rate at which energy is transferred, used, or transformed. Power is measured in units of **watt** (W), which is defined as one joule per second. For example, the rate at which a light bulb transforms electrical energy into heat and light is measured in watts—the more wattage, the more power, or equivalently the more electrical energy is used per unit time. Similarly, **sound power** expresses how much

Table 1.1 Typical intensity values given in W/m^2 (intensity), in decibels (intensity level), and by a factor compared with the TOH.

Source	Intensity	Intensity level	× TOH
Threshold of hearing (TOH)	10^{-12}	**0 dB**	1
Whisper	10^{-10}	**20 dB**	10^{2}
Pianissimo	10^{-8}	**40 dB**	10^{4}
Normal conversation	10^{-6}	**60 dB**	10^{6}
Fortissimo	10^{-2}	**100 dB**	10^{10}
Threshold of pain	10	**130 dB**	10^{13}
Jet take-off	10^{2}	**140 dB**	10^{14}
Instant perforation of eardrum	10^{4}	**160 dB**	10^{16}

energy per unit time is emitted by a sound source passing in all directions through the air. The term **sound intensity** is then used to denote the sound power per unit area.

In practice, sound power and sound intensity can show extremely small values that are still relevant for human listeners. For example, the **threshold of hearing** (TOH), which is the minimum sound intensity of a pure tone a human can hear, is as small as

$$I_{\mathrm{TOH}} := 10^{-12}\ \mathrm{W/m^2}. \tag{1.5}$$

Furthermore, the range of intensities a human can perceive is extremely large with $I_{\mathrm{TOP}} := 10\ \mathrm{W/m^2}$ being the **threshold of pain** (TOP). For practical reasons, one switches to a logarithmic scale to express power and intensity. More precisely, one uses a **decibel** (dB) scale, which is a logarithmic unit expressing the ratio between two values. Typically, one of the values serves as a reference, such as I_{TOH} in the case of sound intensity. Then the intensity measured in dB is defined as

$$\mathrm{dB}(I) := 10 \cdot \log_{10}\left(\frac{I}{I_{\mathrm{TOH}}}\right). \tag{1.6}$$

From this definition, one obtains $\mathrm{dB}(I_{\mathrm{TOH}}) = 0$, and a doubling of the intensity results in an increase of roughly 3 dB:

$$\mathrm{dB}(2 \cdot I) = 10 \cdot \log_{10}(2) + \mathrm{dB}(I) \approx 3 + \mathrm{dB}(I). \tag{1.7}$$

When specifying intensity values in terms of decibels, one also speaks of **intensity levels**. Table 1.1 shows some typical intensity values given in W/m^2 as well as in decibels for some sound sources and dynamics indicators. For example, notes being played pianissimo ("very softly") typically result in intensity levels around 40 dB, whereas notes being played fortissimo ("very loudly") can reach levels up to 100 dB.

We now come back to the concept of **loudness**, which is the perceptual correlate to sound intensity [6, 18]. As said before, the loudness is affected by a number of factors. First of all, the same sound may be perceived to have different loudness depending on the individual. In particular, age is one factor that affects the human ear's response to a sound. Also, the duration of the sound influences perception, since the human auditory system averages the effect of sound intensity over an interval up to a second. Therefore, a human has the feeling that a sound lasting for 200 ms

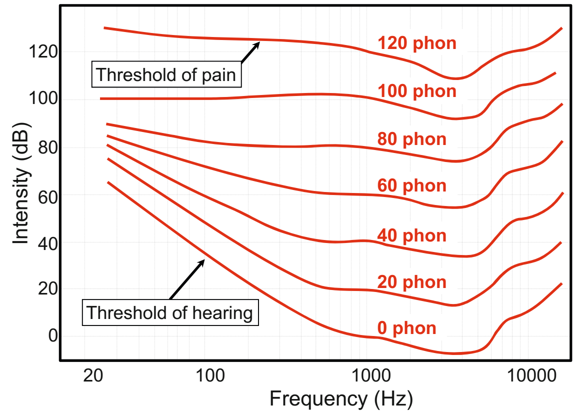


Fig. 1.21 Equal loudness contours (see [6, 18]).

is louder than a similar sound only lasting 50 ms. Furthermore, two sounds with the same intensity but different frequencies are generally not perceived to have the same loudness. Humans with normal hearing are most sensitive to sounds around 2 to 4 kHz, with sensitivity declining for lower as well as higher frequencies. Based on psychoacoustic experiments, the perceived loudness of pure tones depending on the frequency has been determined and expressed by the unit **phon**. Figure 1.21 shows **equal loudness contours**. Each contour line specifies for a fixed loudness given in phons the sound intensities over a (logarithmically spaced) frequency axis. The unit of a phon is normalized with respect to the frequency of $1,000$ Hz, where a phon value equals the intensity level in dB. The contour for 0 phon shows how the threshold of hearing depends on frequency.

1.3.4 Timbre

Besides pitch, loudness, and duration, there is another fundamental aspect of sound referred to as **timbre** or **tone color**. Timbre allows a listener to distinguish the musical tone of a violin, an oboe, or a trumpet even if the tone is played at the same pitch and with the same loudness. As with pitch and loudness, timbre is a perceptual property of sound [28]. However, timbre is very hard to grasp, and because of its vagueness, it is often described in an indirect way: timbre is the attribute whereby a listener can judge two sounds as dissimilar using any criterion other than pitch, loudness, and duration. For example, timbre information allows us to tell apart the sounds produced by the oboe and the violin, even when the pitch and loudness of the sounds are identical [20]. The sound of a musical instrument may be described with such words as bright, dark, warm, harsh, and other terms. Researchers have tried to approach timbre by looking at correlations to more objective sound characteristics such as the temporal and spectral evolution, the absence or presence of tonal and

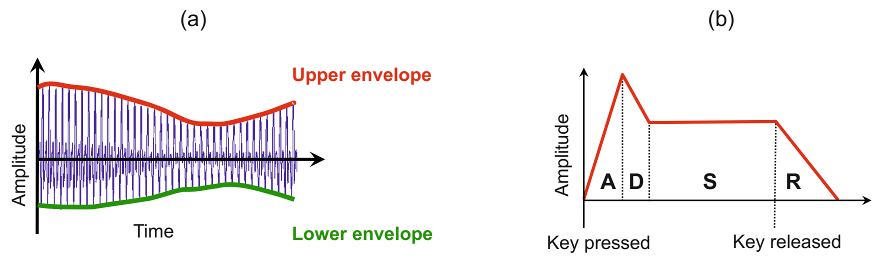


Fig. 1.22 **(a)** Envelope of a signal. **(b)** Schematic view of an ADSR envelope.

noise-like components, or the energy distribution across the partials of a tone. In the following, we take a closer look at some of these characteristics.

When striking a piano key, the resulting sound is much more than a superposition of pure sinusoids that correspond to the fundamental frequency and its overtones. Playing a single note already produces a complex sound mixture with characteristics that may constantly change over time, containing periodic as well as nonperiodic components. At the beginning of a musical tone, there is often a sudden increase of energy. In this short phase, the **attack phase** of the tone, the sound builds up. It contains a high degree of nonperiodic components that are spread over the entire range of frequencies, a property that is also inherent to **noise**. In acoustics, such a noise-like short-duration sound of high amplitude occurring at the beginning of a waveform is also called a **transient**. In the case of a piano, striking a key triggers an entire chain of mechanical actions before a hammer hits one or several strings. All these actions, starting with the finger touching the key and ending with the hammer hitting the strings, produce mechanical noise that merges with the acoustic effects of the strings' excitation. After the attack phase, the sound of a musical tone stabilizes (**decay phase**) and reaches a steady phase with a (more or less) periodic pattern. This third phase, which is also called the **sustain phase**, makes up most of the duration of a musical tone, where the energy remains more or less constant or slightly decreases as is the case with a piano sound. In the final phase of a musical tone, also called the **release phase**, the musical tone fades away. For a piano, this phase starts as soon as the finger leaves the key and the damper stops the strings' vibrations.

Intuitively, the **envelope** of a waveform can be regarded to be a smooth curve outlining its extremes in amplitude (see Figure 1.22a). The different phases as described above have a strong influence on the shape of the envelope of a musical tone. In sound synthesis, the envelope of a signal to be generated is often described by a model called **ADSR**, which consists of an attack (A), decay (D), sustain (S), and release (R) phase (see Figure 1.22b). The relative durations and the amplitudes of the four phases have a significant impact on how the synthesized tone will sound.

The ADSR model is a strong simplification and only yields a meaningful approximation for amplitude envelopes of tones that are generated by certain instruments. For example, the musical tone shown in Figure 1.23a, which is the note C4 played on a piano, has an envelope that is similar to the one suggested by the ADSR model.

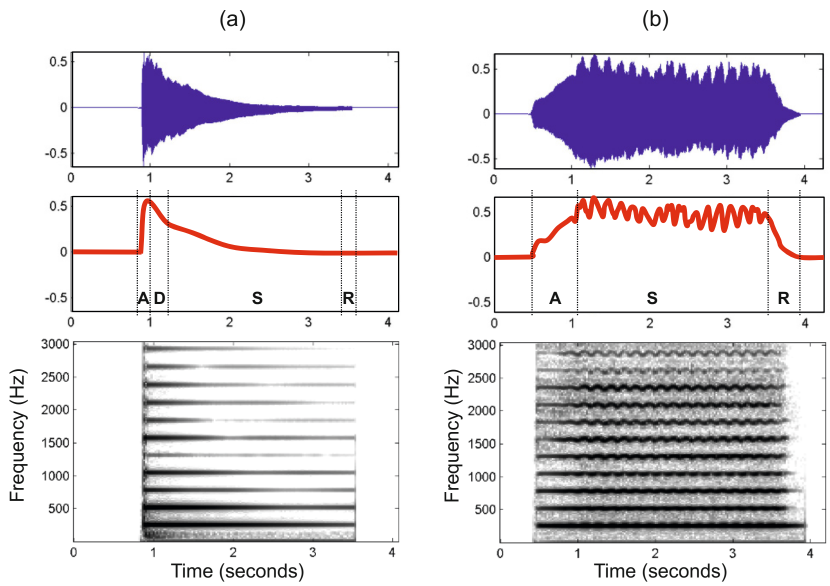


Fig. 1.23 Waveform, amplitude envelope, and spectrogram representation for different instruments playing the same note C4 (261.6 Hz). **(a)** Piano. **(b)** Violin.

After a sharp attack (when the hammer hits the string) and a stabilizing decay, the tone continuously fades out. In the case of a piano sound, the decrease in sound intensity is very slow as long as the damper does not touch the string. Therefore, one can regard this phase as a kind of sustain phase. When the piano key is released and the damper stops the string's vibration, the sound quickly comes to an end. For other instruments, however, the amplitude may evolve in a completely different fashion. This is illustrated by Figure 1.23b, which shows an envelope for the note C4 played on a violin. First of all, since the tone is played softly with a gradual increase in volume, the attack phase is spread out in time. Furthermore, there does not seem to be any decay phase and the subsequent sustain phase is not steady; instead, the amplitude envelope oscillates in a regular fashion. The release phase starts when the violinist stops exciting the string with the bow. The sound then quickly fades out.

For our violin example, one can observe periodic variations in amplitude. This phenomenon, known as **tremolo**, is generated by certain playing styles used for string or wind instruments. The effect of tremolo often goes along with **vibrato**, which is a musical effect consisting of a regular, pulsating change of frequency. Besides string music, vibrato is mainly used by human singers to add expression. In technical terms, tremolo corresponds to an **amplitude modulation**, whereas vibrato corresponds to a **frequency modulation**. Both tremolo and vibrato depend on two parameters: the extent of the variation and the rate at which the amplitude or frequency is varied. Even though tremolo and vibrato are simply local changes in intensity and frequency, they do not necessarily evoke a perceived change in loudness or pitch of the overall musical tone. Rather, they are features that influence the timbre of a musical tone.

Perhaps the most important and well-known property for characterizing timbre is the existence of certain partials and their relative strengths [20]. Recall from Section 1.3.2 that partials are the dominant frequencies of a musical tone with the lowest partial being the fundamental frequency. The inharmonicity expresses the extent to which a partial deviates from the closest ideal harmonic. For harmonic sounds such as a musical tone with a clearly perceivable pitch, most of the partials are close to being harmonics. However, not all partials need to occur with the same strength, as we will see in a moment.

The composition of a sound in terms of its partials can be visualized by a so-called **spectrogram**, which shows the intensity of the occurring frequencies over time. For a detailed introduction on such time–frequency representations refer to Section 2.5. Figure 1.23a shows at the bottom a spectrogram for the note C4 played on a piano, where the intensity is reflected by the shade of gray (the darker the more intense). Both the fundamental frequency of the note (261.6 Hz) as well as its harmonics (integer multiples of 261.6 Hz) are visible as horizontal lines. The decay of the musical tone is reflected by a corresponding decay in each of the partials. Most of the tone's energy is contained in the lower partials, and the energy tends to be lower for the higher partials. Such a distribution is typical for many instruments.

For string instruments, sounds tend to have a rich spectrum of partials, where lots of energy may also be contained in the upper harmonics (see Figure 1.23b). This figure also reveals the vibrato as a regular oscillation in the time–frequency plane. Certain classes of wind instruments including the clarinet (so-called **closed-pipe** wind instruments) produce a very characteristic spectrum of partials. For a cylindrical wind instrument that is open at one end, but closed at the other (at the mouthpiece), one can show that the even harmonics do not show up. In other words, most energy is contained in the odd harmonics ω_0, $3\omega_0$, $5\omega_0$, $\ldots$, with ω_0 denoting the fundamental frequency. For a musical tone played on a bassoon, the fundamental frequency often contains much less energy compared with the higher partials. In contrast, for a tuning fork, most energy is contained in the fundamental frequency, resulting in a sound that is close to a synthesized sinusoid. Instruments such as bells have a very complex spectrum with lots of inharmonicities, which often evokes in the listener the feeling of a bell being out-of-tune. For stringed instruments, one can often measure substantial deviations between higher partials and the theoretical harmonics. The less elastic a string is (that is, the shorter, thicker, higher tension or stiffer it is), the more inharmonicity it may exhibit. This particularly holds for the piano, where such inharmonicities have a crucial influence on the timbre.

With this discussion, we want to indicate that timbre is a multidimensional phenomenon that is hard to measure. It is the irregularities and variations that make a musical tone sound interesting and that give it a particular and natural quality.

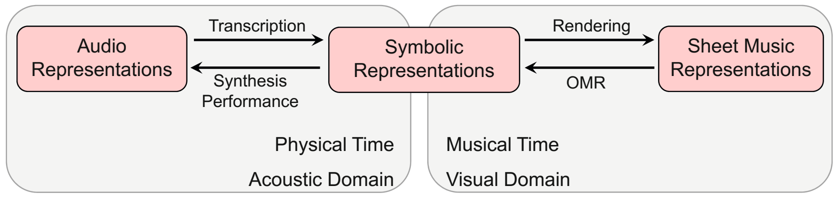


Fig. 1.24 Illustration of three classes of music representation and their relations.

1.4 Further Notes

In this chapter, we looked at three different classes of music representations while introducing some musical and technical terminology that is used throughout this book. We use the term **sheet music** to refer to visual representations of a musical score either given in printed form or encoded digitally in some image format. The term **symbolic** stands for any kind of symbolic representation where the entities have an explicit musical meaning. Finally, the term **audio** is used to denote music recordings given in the form of acoustic waveforms. The boundaries between these classes are not clear. In particular, as illustrated by Figure 1.24, symbolic representations may be close to both sheet music as well as audio representations. On the one hand, symbolic representations such as MusicXML are used for **rendering** sheet music, where the shape of the note objects and their arrangement on a page are determined. As we have seen, optical music recognition (OMR) is the inverse process with the goal of transforming sheet music into a symbolic representation. On the other hand, symbolic representations such as MIDI are used for **synthesizing** audio, where the note objects are transformed into musical tones and real sounds. The inverse process is known as **music transcription**, where the objective is to extract note events, key signature, time signature, instrumentation, and other score parameters from a given music recording [16].

In a sense, symbolic representations can be regarded as the link between the visual (or symbolic) domain accommodating sheet music representations and the acoustic (or physical) domain accommodating audio representations. In the first case, timing is specified in terms of the shape and the relative arrangement of the musical symbols and is typically given in musical units such as measures or beats. In the latter case, timing is specified in physical units such as seconds. For music recordings, there are often no sharp note onsets or offsets (think of a soft onset for a note played on a violin or a gradual fade-out) and the specification of the beginning and the end of musical events becomes an ill-defined problem.

Of course, any kind of categorization of music representations goes along with an oversimplification. Our categorization is far from being comprehensive. We have seen that, when describing musical attributes such as pitch, loudness, and timbre, human perception is a crucial factor. Therefore, besides the acoustic and visual domain, Babbitt [1] considers an additional **auditory** domain. In his taxonomy, a graphemic note (the blob on the page) corresponds in meaning with the (auditory) percept of the note. From a philosophical point of view, as argued by Wiggins et al. [32], music is actually something abstract and intangible, which does not have real existence in itself. In this sense, all of the domain-specific representations are *aspects* of music, but none of them *is* music, individually. Mazzola [17] considers music to be the universe of all different perspectives one may assume. For a psychologically based approach to music along with expectations and emotions it evokes, we refer to the book by Huron [14]. Taking the risk of oversimplification, we adopt in this book a more technically oriented view of music processing and leave out perhaps the most important aspect of music: the human mind.

We now give some references to literature as well as to formats, software, and freely available datasets that may be of interest to the reader. First of all, we want to mention Wikipedia, which is a rich source of useful information on music and audio. Some parts of this chapter are inspired by Wikipedia articles, and we recommend the reader to browse through Wikipedia to further explore the numerous aspects related to music. This chapter is a vastly expanded version of [19, Section 2.1] on music representations, including new material and many new figures. Some of the presented material, especially on sheet music and OMR, was inspired by [8, 9, 30, 31], which discuss techniques for bridging the gap between sheet music and audio representations.

Sheet music has a history of hundreds of years, and the basic concepts we have presented can be found in most introductory textbooks on music notation. Because of significant digitization efforts, sheet music is now widely available in digital formats. In particular for Western classical music, scanned versions of musical editions out of copyright are now freely accessible on the world wide web. One prominent example is the *Petrucci Music Library*, which is a virtual library of public-domain music scores collected within the *International Music Score Library Project* (IMSLP).[3]

For symbolic music, many formats have been suggested in the literature to represent sheet music in a digital, machine-readable form. We refer to [25, 27] for an overview. A comprehensive account on MIDI[4] and its use with electronic instruments and sequencers can be found in [13]. In the book edited by Selfridge-Field [27], one not only finds an introduction to the MIDI format, but also a detailed overview and description of symbolic formats up to the year 1997. Since then, many new formats have been proposed and developed, including both open and well-documented formats, as well as proprietary formats that are bound to specific software packages. The MusicXML[5] format [10, 11] and the MEI[6] format developed by the community-driven Music Encoding Initiative [12, 23], the Hum-

[3] `http://imslp.org`
[4] `http://www.midi.org`
[5] `http://www.musicxml.com`
[6] `http://music-encoding.org/home`

drum[7] format [26], the Lilypond[8] format [21], and the ABC[9] music notation format are prominent examples. The cited articles and websites specified in the footnotes supply further information and links to documentation, applications, and available datasets given in these formats, as well as software.

Several commercial and noncommercial OMR software systems exist. Three popular commercial systems are SharpEye,[10] SmartScore,[11] and PhotoScore.[12] All of them operate on common Western classical music. While the former two only work for printed sheet music, PhotoScore also offers the recognition of handwritten scores. Also, some noncommercial OMR systems such as Gamera[13] and Audiveris[14] exist. Since the introduction of OMR in the late 1960s [22], many researchers have worked on this problem, and we refer to the *Optical Music Recognition Bibliography*[15] for a comprehensive overview of OMR literature. Several studies on the performance of OMR systems and the types of errors that occur have been conducted [2, 3, 4, 8]. Those studies show that OMR systems vary with regard to their strengths and weaknesses. However, the types or classes of recognition errors, as indicated in Figure 1.16, are similar for most systems. Therefore, there is still a need for research for OMR to obtain symbolic music representations of high quality. In this context, the development of top-down strategies that exploit musical knowledge seems to be a promising research direction [24].

For the audio domain, the classic book by Fletcher and Rossing [7] gives a detailed account on musical sound waves and the physics behind their generation by musical instruments. The book by Fastl and Zwicker [5] as well as the one by Moore [18] give deeper insights into the field of auditory perception and psychoacoustics for general audio signals. A series of edited articles on the perception of musical sound patterns has been collected in [15]. The book by Sethares [28] on tuning, timbre, spectrum, and scale provides very interesting explanations of why and how these various aspects are related. This book has been a source of inspiration for this chapter. It also contains many more useful links to the literature, describes insightful experiments, and provides sound examples. A signal-processing-oriented approach to the concepts of timbre and instrumentation can be found in [20].

We finally want to remark that the complexity of an audio representation dramatically increases when moving from simple musical tones to polyphonic (orchestral) music, where the components of various musical tones interfere with each other and intermingle. This renders the task of determining note parameters from the waveform of a polyphonic orchestral piece a difficult one. Except for very sim-

[7] `http://kern.ccarh.org`
[8] `http://www.mutopiaproject.org`
[9] `http://abcnotation.com`
[10] `http://www.music-scanning.com`
[11] `http://www.musitek.com`
[12] `http://www.sibelius.at/photoscore.htm`
[13] `http://gamera.informatik.hsnr.de`
[14] `http://audiveris.kenai.com`
[15] `http://ddmal.music.mcgill.ca`

ple music, the automatic conversion of a music performance into score notation by a computer—a task Mozart was capable of after listening to a polyphonic cantata only once—is still a largely open problem despite decades of research. For a collection of edited articles on this topic, we refer to [16].

References

1. M. Babbitt, *The use of computers in musicological research*, Perspectives of New Music, 3 (1965), pp. 74–83.
2. E. P. Bugge, K. L. Juncher, B. S. Mathiasen, and J. G. Simonsen, *Using sequence alignment and voting to improve optical music recognition from multiple recognizers*, in Proceedings of the International Society for Music Information Retrieval Conference (ISMIR), Miami, Florida, USA, 2011, pp. 405–410.
3. D. Byrd, W. Guerin, M. Schindele, and I. Knopke, *OMR evaluation and prospects for improved OMR via multiple recognizers*, technical report, Indiana University, Bloomington, Indiana, USA, 2010.
4. D. Byrd and M. Schindele, *Prospects for improving OMR with multiple recognizers*, in Proceedings of the International Society for Music Information Retrieval Conference (ISMIR), Victoria, Canada, 2006, pp. 41–46.
5. H. Fastl and E. Zwicker, *Psychoacoustics, Facts and Models*, Springer, 3rd ed., 2007.
6. H. Fletcher and W. A. Munson, *Loudness, its definition, measurement and calculation*, Journal of the Acoustic Society of America, 5 (1933), pp. 82–108.
7. N. H. Fletcher and T. D. Rossing, *The Physics of Musical Instruments*, Springer-Verlag, 1991.
8. C. Fremerey, *Automatic Organization of Digital Music Documents – Sheet Music and Audio*, PhD thesis, University of Bonn, 2010.
9. C. Fremerey, M. Müller, and M. Clausen, *Towards bridging the gap between sheet music and audio*, in Knowledge Representation for Intelligent Music Processing, E. Selfridge-Field, F. Wiering, and G. A. Wiggins, eds., Dagstuhl Seminar Proceedings, Dagstuhl, Germany, 2009, Leibniz-Zentrum für Informatik.
10. M. Good, *MusicXML for notation and analysis*, Computing in Musicology, 12 (2001), pp. 113–124.
11. ———, *Lessons from the adoption of MusicXML as an interchange standard*, in Proceedings of XML, 2006.
12. A. Hankinson, P. Roland, and I. Fujinaga, *The music encoding initiative as a document-encoding framework*, in Proceedings of the International Society for Music Information Retrieval Conference (ISMIR), Miami, Florida, USA, 2011, pp. 293–298.
13. D. M. Huber, *The MIDI manual*, Focal Press, 3rd ed., 2006.
14. D. B. Huron, *Sweet anticipation: Music and the psychology of expectation*, The MIT Press, 2006.
15. M. R. Jones, R. R. Fay, and A. N. Popper, eds., *Music Perception (Springer Handbook of Auditory Research)*, Springer, 2010.
16. A. P. Klapuri and M. Davy, eds., *Signal Processing Methods for Music Transcription*, Springer, New York, 2006.
17. G. Mazzola, *The topos of music*, Birkhäuser, 2002.
18. B. C. Moore, *An Introduction to the Psychology of Hearing*, Emerald/Brill, 6th ed., 2012.
19. M. Müller, *Information Retrieval for Music and Motion*, Springer Verlag, 2007.
20. M. Müller and A. Klapuri, *Music Signal Processing*, vol. 4 of Library in Signal Processing, Academic Press, 2013, ch. 27, pp. 713–756.
21. H.-W. Nienhuys and J. Nieuwenhuizen, *Lilypond, a system for automated music engraving*, in Proceedings of the XIV Colloquium on Musical Informatics, Firenze, Italy, 2003, pp. 664–665.
22. D. H. Pruslin, *Automatic recognition of sheet music*, PhD thesis, Massachusetts Institute of Technology, 1966.
23. L. Pugin, J. Kepper, P. Roland, M. Hartwig, and A. Hankinson, *Separating presentation and content in MEI*, in Proceedings of the International Society for Music Information Retrieval Conference (ISMIR), Porto, Portugal, 2012, pp. 505–510.
24. C. Raphael and J. Wang, *New approaches to optical music recognition*, in Proceedings of the International Society for Music Information Retrieval Conference (ISMIR), Miami, Florida, USA, 2011, pp. 305–310.

25. C. ROADS, *The computer music tutorial*, The MIT Press, 1996.
26. C. S. SAPP, *Online database of scores in the humdrum file format*, in Proceedings of the International Society for Music Information Retrieval Conference (ISMIR), London, UK, 2005, pp. 664–665.
27. E. SELFRIDGE-FIELD, ed., *Beyond MIDI: the handbook of musical codes*, MIT Press, Cambridge, Massachusetts, USA, 1997.
28. W. A. SETHARES, *Tuning, Timbre, Spectrum, Scale*, Springer, London, 1998.
29. R. N. SHEPARD, *Circularity in judgments of relative pitch*, Journal of the Acoustic Society of America, 36 (1964), pp. 2346–2353.
30. V. THOMAS, *Music Synchronization, Audio Matching, Pattern Detection, and User Interfaces for a Digital Music Library System*, PhD thesis, University of Bonn, 2013.
31. V. THOMAS, C. FREMEREY, M. MÜLLER, AND M. CLAUSEN, *Linking sheet music and audio - challenges and new approaches*, in Multimodal Music Processing, M. Müller, M. Goto, and M. Schedl, eds., vol. 3 of Dagstuhl Follow-Ups, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 2012, pp. 1–22.
32. G. WIGGINS, D. MÜLLENSIEFEN, AND M. PEARCE, *On the non-existence of music: Why music theory is a figment of the imagination*, Musicae Scientiae, Discussion Forum, 5 (2010), pp. 231–255.